



2026 BASIC ABSTRACT ALGEBRA STUDY

# LECTURE 1 HOMEWORKS

## SOLUTION - Q2

**Version**

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**Instructor**

Abiria

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**Read this carefully before moving on.**

Please, clearly specify whether the statement in question is *true* or *false*, for the *prove* or *disprove* type of questions. As always, it is highly encouraged for you to write each step clearly with full sentences.

As a demonstration of a proof answer, consider the following example.

The above statement is **true**.

*Proof.* ...



As a demonstration of a disproof answer, consider the following example.

The above statement is **false**.

*Proof.* ...

**Note 1 (Notations)**

The following notations will be used throughout the following questions.

- $\mathbb{Z}$ : A set of all integers.
- $\mathbb{Z}^+$ : A set of all positive integers. i.e.,  $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$ .
- $\mathbb{Z}_n$ : A set of non-negative integers from 0 to  $n - 1$  for some positive integer  $n$ . For instance,  $\mathbb{Z}_4 = \{0, 1, 2, 3\}$ .
- $n\mathbb{Z}$ : A set given as  $\{nx \mid x \in \mathbb{Z}\}$  for an integer  $n$ . i.e., the set of all multiples of  $n$  in  $\mathbb{Z}$ . For instance,  $5\mathbb{Z} = \{\dots, -10, -5, 0, 5, 10, \dots\}$ .
- $\mathbb{Q}$ : A set of all rational numbers.
- $\mathbb{R}$ : A set of all real numbers.
- $\mathbb{R}^+$ : A set of all positive real numbers.
- $\mathbb{C}$ : A set of all complex numbers.
- $A + B$ : A set given as  $\{a + b \mid a \in A, b \in B\}$ . For instance,  $\mathbb{Z}_2 + \mathbb{Z}_3 = \{(0 + 0), (0 + 1), (0 + 2), (1 + 0), (1 + 1), (1 + 2)\} = \{0, 1, 2, 3\}$

**Definition 2 (Subset)**

Given two sets  $A$  and  $B$ ,  $A$  is said to be a subset of  $B$  if the following condition holds.

If  $x$  is an element of  $A$ , then  $x$  is an element of  $B$ .

**Definition 3 (Extensionality)**

Two sets  $A$  and  $B$  are said to be equal if and only if  $A$  is a subset of  $B$  and  $B$  is a subset of  $A$ . This is called the *axiom of extensionality*. If two sets  $A$  and  $B$  are equal, we write  $A = B$ .

**Q1.**

**EASY**

Let  $A$  and  $B$  be two sets. Let  $A(x)$  mean  $x$  is an element of  $A$ , and  $B(x)$  mean  $x$  is an element of  $B$ .

Which of the following is equivalent to saying that  $A \subset B$ ?

- ☐ 1.  $B(x) \rightarrow A(x)$ 
☐ 2.  $\text{not } A(x) \rightarrow B(x)$ 
☐ 3.  $\text{not } A(x) \rightarrow \text{not } B(x)$
- ☐ 4.  $A(x) \rightarrow \text{not } B(x)$ 
☐ 5.  $\text{not } B(x) \rightarrow \text{not } A(x)$

#### Definition 4 (Roots of unity)

Let  $\mu_n$  denote the set of all complex solutions to an equation of form  $x^n = 1$  for a positive integer  $n$ . Elements of such a set are called  $n$ -th roots of unity.

For instance,  $\mu_2 = \{1, -1\}$ .

Q2.

EASY

Show that  $\mu_2 \subset \mu_4$ .

*Proof.* Let's verify that by manually listing elements of  $\mu_2$  and  $\mu_4$ .

There are two complex solutions to  $x^2 = 1$ : 1 and  $-1$ .

Meanwhile, there are four complex solutions to  $x^4 = 1$ : 1,  $-1$ ,  $i$ , and  $-i$ .

From here, clearly  $\mu_2 \subset \mu_4$ . □

Now, let's consider squaring each element of  $\mu_2$ . By the definition,  $1^2 = 1$  and  $(-1)^2 = 1$ . Again, let's square them once more.  $(1^2)^2 = 1^2 = 1$ , and similarly  $((-1)^2)^2 = 1^2 = 1$ . Can you find a pattern?

Q3.

EASY

Prove or disprove that  $\mathbb{Z} \subset n\mathbb{Z} + m\mathbb{Z}$  for any two distinct positive integers  $n$  and  $m$ .

**Hint 5.** Provide concrete  $n$  and  $m$  that make up a counterexample. Try justifying your answer by using the definition of set builder notation.

Q4.

HARD

Prove or disprove that  $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$ .

**Hint 6.** cf. lecture note.

Q5.

HARD

Show using [Definition 2](#) that for any two positive integers  $n$  and  $m$ ,  $\mu_n \subset \mu_{nm}$ .

**Hint 7.** Of course, it is a generalization of [Q2](#).

HARD

Q6.

Show by [Definition 3](#) that  $\mathbb{Z}_n + \mathbb{Z}_n = \mathbb{Z}_{2n-1}$  for any positive integer  $n$ .

**Hint 8.** cf. PCSA Q4.

HARD

Q7.

Prove or disprove that if  $\mathbb{Z} = n\mathbb{Z}$  for an integer  $n$ , then either  $n = 1$  or  $n = -1$ .

**Hint 9.** What are the factors of 1 and  $-1$ ?

### Definition 10 (Symmetric difference)

$A \triangle B$  is a binary set operation given by  $(A \setminus B) \cup (B \setminus A)$  and is called a *symmetric difference* between  $A$  and  $B$ .

Q8.

LEVI

Problem by @escentia07.

Prove or disprove that  $A \triangle B \subset C$  if and only if  $A \cup B \subset (A \cap B) \cup C$ .

Q9.

LEVI

For an integer  $n$ , let  $M(k)$  denote the set  $\{x \mid x \equiv k \pmod{n}\}$ . Let  $P$  be the set given by  $P = \{M(k) \mid k \in \mathbb{Z}\}$ .

(a) Show that every two distinct elements of  $P$  are disjoint.

**Hint 11.** Use [Definition 3](#) and contraposition.

(b) Show that the following holds for any two integers  $a$  and  $b$  by [Definition 3](#).

$$M(a) + M(b) = M(a + b)$$

Q10.

LEVI

Problem by @escentia07.

Let  $f(x) = x^{2121} + x - 1$

How many real solutions does the equation  $f^{2026}(x) = x$  have? Justify your answer.

Answer: \_\_\_\_\_

Q11.

LEVI

Let  $T$  be the set  $\{z \in \mathbb{C} \mid |z| = 1\}$  and  $\mu_\infty$  be the set of all roots of unity as follows.

$$\mu_\infty = \bigcup_{n=1}^{\infty} \mu_n = \mu_1 \cup \mu_2 \cup \dots$$

Prove or disprove that  $T = \mu_\infty$ .

End.

You may now submit your work.

## Optional evaluation sheet

Unlike PCSA, submitting the sheet is **optional** for homeworks.

For each question so far, you can evaluate it based on the following criteria:

- If you failed to understand the question, mark  $\times$  in the Evaluation column.
- If you understood the question, but didn't know how to solve it, mark  $?$  in the Evaluation column.
- If you understood the question and are sure you have solved it, mark  $\circ$  in the Evaluation column.

| No. | Evaluation | No. | Evaluation |
|-----|------------|-----|------------|
| Q1  |            | Q7  |            |
| Q2  |            | Q8  |            |
| Q3  |            | Q9  |            |
| Q4  |            | Q10 |            |
| Q5  |            | Q11 |            |
| Q6  |            |     |            |

There are these additional questions as well; feel free to answer them.

- (a) Which was the hardest for you, among those you have solved? \_\_\_\_\_
- (b) Which seemed to be the hardest for you, among those you haven't solved? \_\_\_\_\_
- (c) Which was your favorite question, if you had to choose one? \_\_\_\_\_